

General Ozochiawaeze

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Education

- 2021 – Present  **Ph.D. Candidate in Mathematics, Purdue University**
Expected graduation: Spring 2027.
Research interests: Applied Analysis, PDEs and Inverse Problems.
Advisors: Peijun Li and Isaac Harris.
- 2021  **M.S. in Applied Mathematics, New Jersey Institute of Technology**
Advisor: Christina Frederick.
- 2018  **B.A. in Mathematics, Rutgers University.**

Research Publications

Journal Articles

- 1 R. Ceja Ayala, I. Harris, and G. Ozochiawaeze, “Factorization method for the biharmonic scattering problem for an absorbing penetrable scatterer,” *Communications on Analysis and Computation*, vol. 9, pp. 16–39, 2026.  DOI: 10.3934/cac.2026010
- 2 I. Harris, P. Li, and G. Ozochiawaeze, “Sampling methods for the inverse cavity scattering problem of biharmonic waves,” *Inverse Problems*, vol. 42, p. 015 002, 2026.  DOI: 10.1088/1361-6420/ae2ef8

Preprints

- 1 I. Harris and G. Ozochiawaeze, *Novel implementation of the extended sampling method for inverse biharmonic scattering*, 2026. arXiv: 2605.12367 [math.AP].  URL: <https://arxiv.org/abs/2605.12367>
- 2 G. Ozochiawaeze and I. Harris, *Factorization method for a clamped obstacle from near-field measurements via a far-field transformation*, 2026. arXiv: 2609.20528 [math.AP].  URL: <https://arxiv.org/abs/2609.20528>

Employment History

- May – Oct 2023  **Group 36 Summer Research Intern, MIT Lincoln Laboratory**
- May – Aug 2022  **Group 37 Summer Research Intern, MIT Lincoln Laboratory**
- Jul 2020 – May 2021  **Underwater Acoustic Simulation Researcher, New Jersey Institute of Technology**

Teaching Experience

- Fall 2025  **Lecturer**, Purdue University
MA 16020: Applied Calculus II.
- Spring 2024  **Recitation Instructor**, Purdue University
MA 261: Multivariable Calculus.
- Spring 2023  **Teaching Assistant**, Purdue University
MA 511: Linear Algebra and Applications.
- Fall 2022  **Recitation Instructor**, Purdue University
MA 261: Multivariable Calculus.

Teaching Experience (continued)

Spring 2022	■ Recitation Instructor , Purdue University MA 162: Calculus II.
Fall 2021	■ Recitation Instructor , Purdue University MA 161: Calculus I.
Spring–Fall 2020	■ Instructor , Accel Learning Mentored students preparing for AMC and AIME mathematics competitions.
Fall 2018–Spring 2019	■ Mathematics Instructor , Mathnasium K–12 mathematics teaching and tutoring.

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Conferences & Presentations

Invited and Contributed Talks

2026	■ SIAM Great Lakes Section , Purdue University (Invited Speaker) <i>A Factorization Method for Clamped Obstacles from Near-Field Data</i>
	■ Theory and Applications of Elliptic PDE , UC Irvine (Invited Speaker) <i>Extended Sampling Method for Inverse Biharmonic Scattering</i>
	■ Baby Inverse Problems (BIP) Seminar , Online (Invited Speaker) <i>A Factorization Method Approach to the Biharmonic Transmission Problem in Absorbing Media</i>
2025	■ SIAM Conference on Analysis of PDEs (PD25) , Pittsburgh (Invited Speaker) <i>A Factorization Method Approach to the Biharmonic Transmission Problem in Absorbing Media</i>
	■ SIAM Conference on Analysis of PDEs (PD25) , Pittsburgh (Invited Speaker) <i>Localizing Cavities in Biharmonic Scattering with Limited Data</i>
	■ UNCG PDE Conference , Greensboro, NC <i>Strengthened Analysis of Linear Sampling Method for Biharmonic Wave Scattering</i>
	■ Summer School: Elliptic PDE , UC Irvine <i>Localizing Cavities in Biharmonic Scattering with Limited Data</i>
2023	■ MIT Lincoln Laboratory , Group 36 <i>Compressive Radar Imaging in Missile Defense</i>
2022	■ MIT Lincoln Laboratory , Group 37 <i>Quantifying Real Operational Performances of SONAR Arrays in Random Ocean Environments</i>

Institutional Seminars

2025	■ CCAM Lunch Seminar , Purdue University <i>Localizing Cavities in Biharmonic Scattering with Limited Data</i>
	■ Graduate Student Analysis Seminar , Purdue University <i>Qualitative Reconstruction Methods in Inverse Biharmonic Scattering</i>
2024	■ Purdue Graduate Research Day and SIAM Student Chapter , Purdue University <i>A Linear Sampling Method for Recovering a Clamped Cavity in a Thin Plate</i>
2022	■ Purdue Math Student Colloquium , Purdue University <i>Finite Element Modeling of Underwater Acoustic Environments</i>

Conference Attendance

2026	■ May Midwestern Microlocal Meeting , Chicago, IL
	■ Geometric Spectral Theory and Applications , Isaac Newton Institute, UK

Conferences & Presentations (continued)

- 2025  **Mathematics of Wave Phenomena**, Karlsruhe Institute of Technology, Germany
  **Mathematics and Statistics in Industry Workshop**, Indiana, USA (Panel Discussion)
- 2024  **Paris-Saclay Conference in Analysis and PDE**, Orsay, France
  **Triangle Area Inverse Problems Weekend**, NC State University, Raleigh, NC
- 2022  **New Ideas in Computational Inverse Problems**, BIRS, Banff, Canada
  **Workshop on Scientific Computing**, Purdue University, Indiana

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Skills

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| Computing |  Python, C++, MATLAB. |
| Scientific Software |  COMSOL, FEniCS, PyTorch. |
| Signal Processing |  Array signal processing, beamforming, direction-of-arrival estimation (DOA), adaptive filtering, matched filtering, compressive sensing, sonar and radar applications. |
| Typesetting |  $\text{\LaTeX}$. |

Miscellaneous Experience

Awards and Honors

- 2026  **Bilsland Dissertation Fellowship**
- 2025  **SIAM Student Travel Award**
- 2024  **Special Purdue Employee Recognition**
- 2022  **GEM Fellowship**
- 2020  **Graduate C/Startup Research Grant**
- 2016  **Tau Sigma Honor Society**

Service

- 2026  **Mentor, Directed Reading Program in Markov Chains and Stochastic Processes**, Purdue University. Mentee: Caleb Dai.
- 2025–2026  **Purdue Math Department Graduate Representative**, Purdue University.
- 2025  **Mentor, Directed Reading Program in Computational Topology**, Purdue University. Mentee: Aaron (Sang Hyun) Kim.
- 2022–2025  **Founder and Organizer, Math History Seminar**, Purdue University.
- 2017–2018  **Treasurer and Web Manager, Rutgers Undergraduate Mathematics Association**, Rutgers University.

References

Available on Request